Year 12 Further Maths - Complex numbers

Similar questions - Equations, conjugates and reflection

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

(a) Solve $3w - 2 = \mathrm{i}(4 + 3w)$, giving your answer in the form $a + b\mathrm{i}$. **[4]**

(b) Solve $w - 3\mathrm{i} = 2w^* + 1$. **[4]**

2

(a) Prove algebraically that, for non-zero z, $z = -z^*$ if and only if z is purely imaginary. **[3]**

(b) The complex numbers z and z^* are represented on an Argand diagram by A and B. (i) State the single transformation that maps A to B. (ii) Use a geometric argument to prove that $z = z^*$ if and only if z is purely real. **[3]**

Question 1

(a) $3w-2=4\mathrm{i}+3\mathrm{i}w$, so $w(3-3\mathrm{i})=2+4\mathrm{i}$ (taking the terms in w to one side). $w=\dfrac{2+4\mathrm{i}}{3-3\mathrm{i}}=\dfrac{(2+4\mathrm{i})(3+3\mathrm{i})}{18}=\dfrac{-6+18\mathrm{i}}{18}=-\frac{1}{3}+\mathrm{i}$

(b) Let $w=x+y\mathrm{i}$: $x+(y-3)\mathrm{i}=2x+1-2y\mathrm{i}$. Real: $x=2x+1\Rightarrow x=-1$. Imaginary: $y-3=-2y\Rightarrow y=1$. $w=-1+\mathrm{i}$

Question 2

(a) Let $z=x+y\mathrm{i}$. $z=-z^*\Leftrightarrow x+y\mathrm{i}=-x+y\mathrm{i}\Leftrightarrow x=0$, i.e. z is purely imaginary (and non-zero when $y\neq 0$)

(b)(i) Reflection in the real axis

(ii) $z=z^*$ means A and B coincide. A point is mapped to itself by reflection in the real axis if and only if it lies on the real axis (the line of invariant points), i.e. z is purely real